Sortify: AI-Powered Waste Identification and Disposal Guidance

Project Proposal for UCS503P
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1 Higher-order goal

This project aims to make everyday waste segregation simpler and more accessible by providing users with an intelligent mobile assistant for identifying waste items and determining appropriate disposal methods. While waste management involves broader environmental and infrastructural challenges, the immediate engineering objective is to reduce uncertainty at the point where a user decides how to dispose of an item.

The proposed system, **Sortify**, combines a mobile application, backend services, a waste-information database, and a lightweight machine learning model to provide actionable disposal guidance rather than merely classifying an image.

2 Time-to-value

- The core workflow will focus on delivering immediate value: capture or upload an image, identify the waste item, and provide clear disposal guidance.
- The system will prioritize a reliable and understandable user experience over unnecessary model complexity or research-oriented algorithms.
- The application will be developed as a modular software system so that additional features such as user history, analytics, feedback, and administrative tools can be added without redesigning the core workflow.
- Success will be evaluated through both software engineering quality and the usefulness and reliability of the waste-identification workflow.

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3 Problem Statement

People frequently encounter everyday items whose correct disposal method is unclear. Items such as plastic packaging, batteries, electronic accessories, food waste, medicine strips, and contaminated containers may belong to different waste streams, and the correct handling procedure is not always obvious to users.

Common pain points include:

- **Identification uncertainty:** users may not know which waste category an item belongs to.
- **Fragmented information:** disposal instructions are often scattered across different sources and require manual searching.
- **Incorrect segregation:** uncertainty can result in recyclable, organic, hazardous, or e-waste items being placed in inappropriate waste streams.
- **Lack of actionable guidance:** identifying an item is not sufficient; users need to know what to actually do with it.
- **Location dependence:** disposal procedures can vary according to local waste-management practices and available facilities.

Why this matters: A simple and accessible decision-support system can reduce the effort required to make informed disposal decisions and encourage better waste-segregation habits.

4 Proposed Solution

4.1 Overview

Build a **mobile-first** waste identification and disposal guidance application named **Sortify**. The application will use machine learning as one component of a larger software system.

The proposed solution will contain the following components:

- **Mobile application:** users capture an image using their device camera or upload an existing image.
- **ML-based identification:** a lightweight image-classification model predicts the likely waste item and provides a confidence score.
- **Waste knowledge base:** the predicted item is mapped to its waste category, material information, disposal method, and relevant safety instructions.
- **Location-aware guidance:** disposal recommendations can be associated with the user's selected geographical context so that guidance can account for local practices.
- **Manual search and fallback:** users can search for waste items manually or select a category when the ML model is uncertain.
- **User history:** authenticated users can save scans and view their previous classifications.
- **Feedback system:** users can indicate when a prediction is incorrect, creating structured feedback for future model improvement.
- **Administrative dashboard:** authorized administrators can manage waste items, disposal guidelines, and user feedback.

- **Conversational assistant (chatbot):** an integrated chatbot lets users ask natural-language questions about waste disposal, clarify ambiguous items, and get guidance without manually searching the knowledge base.

4.2 System Context Diagram

Figure 1 provides a high-level view of Sortify and its interaction with the primary external actors.

Figure 1: Level-0 Context Diagram for Sortify

4.3 Core workflow

1. The user captures or uploads an image of a waste item.
2. The application validates the image and sends it to the backend prediction service.
3. The ML model identifies the most likely waste item and returns a confidence score.
4. The backend retrieves the corresponding waste category and disposal information from the knowledge base.
5. The user receives an explanation containing the predicted item, category, confidence, disposal instructions, and relevant safety information.
6. The user can save the result, correct the prediction, or manually search for another item.
7. The scan and any user feedback are stored for history, analytics, and future improvement.

4.4 Use Case Diagram

Figure 2 illustrates the major functional interactions between Sortify users, administrators, and the system.

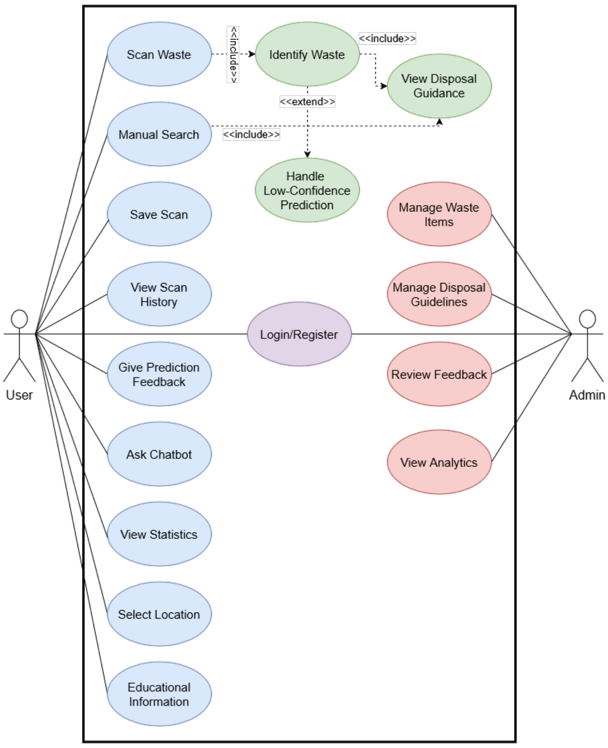

Figure 2: Use Case Diagram for Sortify

4.5 Level-1 Data Flow Diagram

Figure 3 expands the system context into the main functional processes and data interactions within Sortify.

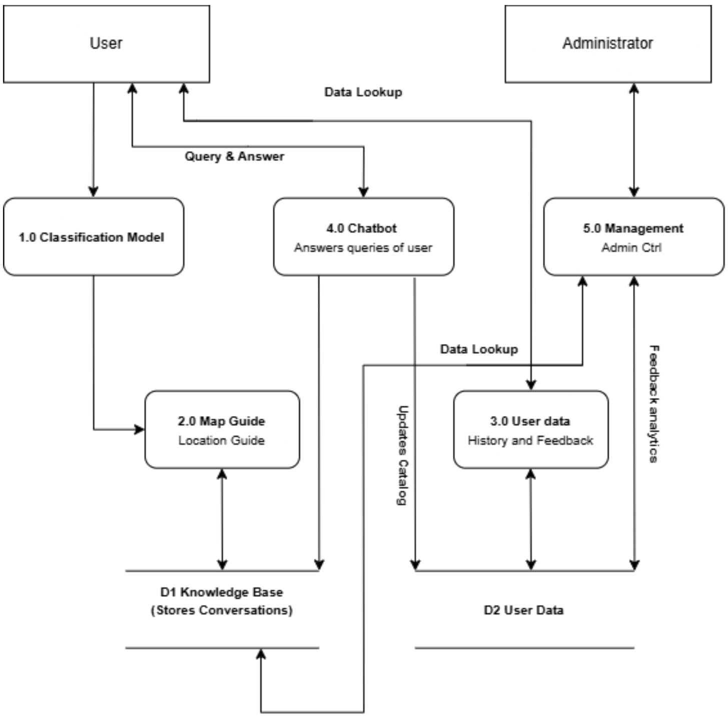

Figure 3: Level-1 Data Flow Diagram for Sortify

4.6 Waste classification structure

The initial system will focus on a limited number of practical waste categories rather than attempting to classify every possible object.

The proposed categories are:

- **Recyclable:** examples include plastic bottles, paper, cardboard, glass, and metal containers.
- **Organic:** examples include food scraps, fruit and vegetable waste, and other biodegradable material.
- **General waste:** items that cannot reasonably be recycled or composted through the supported disposal streams.
- **E-waste:** electronic devices, cables, chargers, and other electronic components.
- **Hazardous waste:** batteries, certain medicines, chemical containers, and other materials requiring special handling.

The ML component may identify a more specific item, such as a "plastic bottle" or "battery", while the application maps that item to the appropriate disposal category and guidance.

5 Solution Approach

This is a Software Engineering project, so the focus will be on building a complete, maintainable, and deployable software system rather than developing a novel research model.

5.1 Machine learning component

The ML component will use a pretrained image-classification architecture with transfer learning or fine-tuning on an appropriate public waste-image dataset.

The model development process will include:

- Dataset preparation and cleaning.
- Image preprocessing and augmentation.
- Training, validation, and testing.
- Evaluation using accuracy, precision, recall, F1-score, and a confusion matrix.
- Confidence-based handling of uncertain predictions.
- Integration of the trained model with the application backend.

The model will not be treated as infallible. Low-confidence predictions will allow the user to retake the image, select an alternative prediction, or use manual search.

Figure 4 summarizes the main processing stages of the classification component.

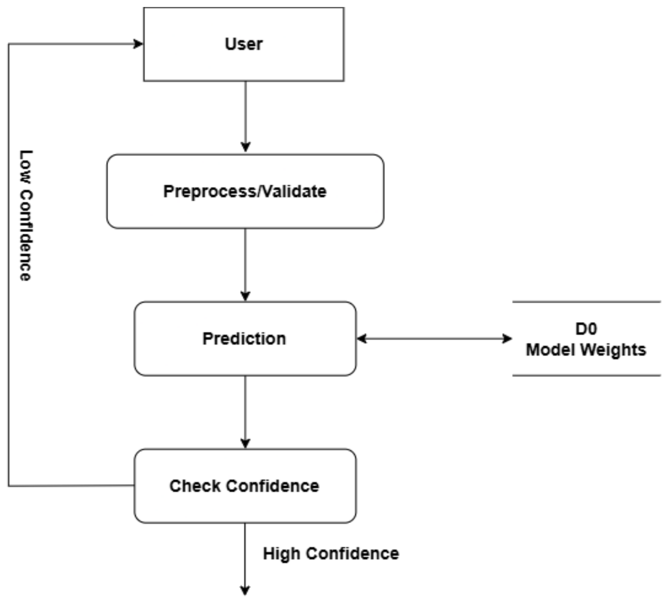

Figure 4: ML Classification Process

5.2 Mobile application

The primary client will be developed using **React Native**. The application will provide:

- Camera and image-upload functionality.
- Prediction and disposal-guidance screens.
- User authentication and profile management.
- Scan history and search.
- Personal statistics and activity dashboard.
- Prediction feedback and correction.
- Educational information about waste segregation.
- Chatbot interface for natural-language disposal queries.

5.3 Backend architecture

The backend will be implemented using **FastAPI** and will expose REST APIs for:

- Authentication and user management.
- Image upload and ML inference.
- Waste-item and disposal-guideline retrieval.
- Scan history.
- User feedback.
- Dashboard and analytics data.
- Administrative operations.

- Chatbot query handling and conversation storage.

A relational database such as PostgreSQL will store users, waste items, guidelines, scans, prediction results, feedback, and relevant administrative data.

Figure 5 illustrates the main user-facing data operations involving saved scans, feedback, and scan history.

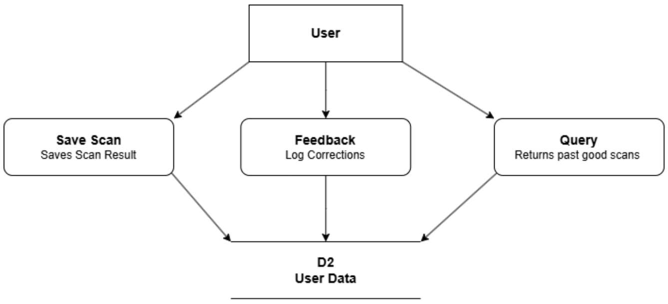

Figure 5: User Data and Feedback Flow

Figure 6 illustrates the corresponding administrative operations for managing the catalog and reviewing feedback and analytics.

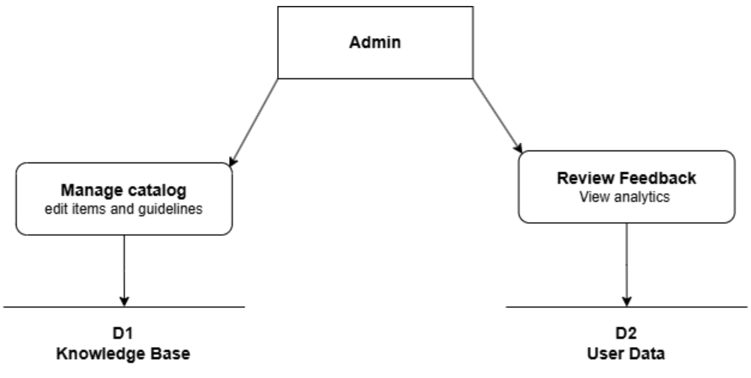

Figure 6: Administrative Management Flow

5.4 Knowledge base and disposal guidance

The application will maintain a structured database of waste items and disposal guidance. Each supported waste item can contain:

- Waste-item name and description.
- Waste category.
- Material information.
- Recyclability or compostability information where applicable.
- Disposal method.
- Preparation instructions.
- Safety warnings.
- Location-specific guidance where available.

This separation between ML prediction and disposal guidance ensures that the model is responsible for recognition while the software system remains responsible for interpreting the prediction and presenting actionable instructions.

Figure 7 shows how waste information and location context contribute to disposal guidance.

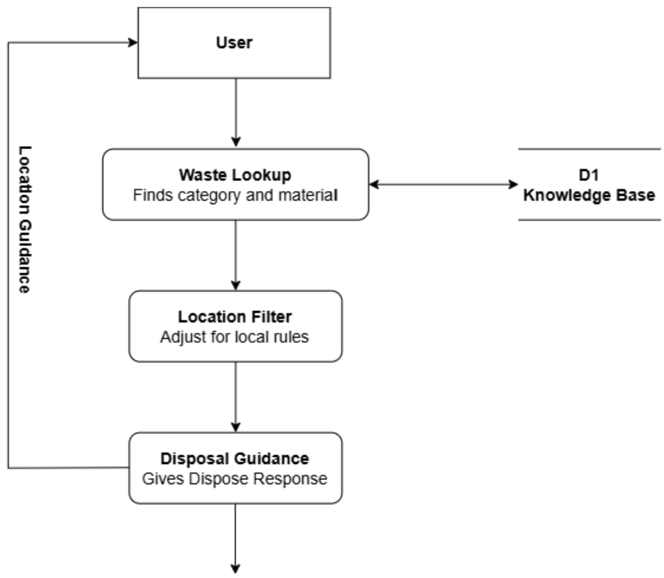

Figure 7: Location-Aware Disposal Guidance Flow

5.5 Conversational assistant

In addition to the image-based scan-to-guidance workflow, Sortify will provide a conversational assistant that lets users interact with the waste knowledge base through natural language, for example by asking how to dispose of a specific item or whether something is recyclable.

- **Grounded responses:** the chatbot will retrieve relevant entries from the waste knowledge base (retrieval-augmented generation) rather than relying on an unconstrained language model, reducing the risk of incorrect disposal advice.
- **Fallback and escalation:** when the assistant cannot confidently answer a query, it will direct the user to manual search or suggest capturing an image for ML-based identification.
- **Integration with scan results:** after a scan, users can ask follow-up questions about the identified item directly within the chatbot interface.
- **Conversation history:** authenticated users' chatbot conversations can be stored alongside scan history for continuity across sessions.
- **Scope control:** the assistant will be restricted to waste-management and disposal-related topics to keep responses relevant and reliable.

Figure 8 summarizes the chatbot's query and response flow.

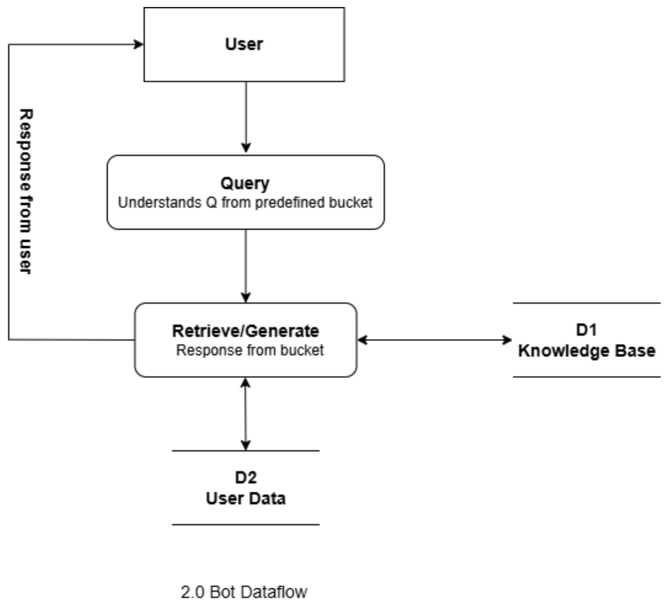

Figure 8: Conversational Assistant Flow

5.6 CI/CD and delivery

CI/CD will be used to:

- Automatically run unit and integration tests on code changes.
- Perform linting and build checks.
- Produce repeatable deployment artifacts.
- Enable controlled deployment of the backend and application services.
- Reduce regression risk as new features are introduced.

6 Evaluation Criterion

The project will evaluate both the software system and the ML component.

6.1 Primary evaluation criteria

End-to-end task success: percentage of test users who can successfully identify an item and reach an appropriate disposal recommendation using the application.

- Measure the complete workflow from image capture/upload to disposal guidance.
- Record failures caused by invalid input, low-confidence predictions, unavailable guidance, or backend errors.

ML classification performance: evaluate the image-classification model on a held-out test set using standard classification metrics.

6.2 Secondary evaluation criteria

- **Accuracy and F1-score:** measure classification performance across supported waste classes.
- **Confidence reliability:** evaluate how effectively the system identifies uncertain predictions and avoids presenting low-confidence results as certain.
- **API reliability:** measure successful request completion and error rates during testing.
- **Usability:** collect user feedback on how easily users can complete the scan-to-disposal workflow.
- **Response time:** measure the time required to process an image and return a result under normal deployment conditions.

6.3 Validation plan

- Test the application using a defined set of representative waste items.
- Evaluate the ML model on a held-out test dataset.
- Conduct functional testing of authentication, image upload, prediction, history, feedback, and administrative workflows.
- Conduct usability testing with representative student users.
- Record and analyze incorrect predictions and user corrections to identify areas for improvement.

7 Scalability

The system should be capable of expanding beyond the initial prototype without requiring a complete architectural redesign.

1. **Scalable architecture:** the mobile frontend, backend API, database, and ML inference components will remain logically separated.
   - Stateless API design where appropriate.
   - Pagination and indexing for scan history and administrative queries.
   - Separation of ML inference from the main application logic.
   - Modular database structures for adding new waste items and locations.
2. **Feature scalability:** the architecture should allow future additions such as:
   - Additional waste categories and waste items.
   - More detailed local disposal guidelines.
   - Collection-point discovery.
   - Barcode-based product identification.
   - Improved ML models.
   - Optional on-device/offline inference.
3. **Operational deployability:** the backend and database should be deployable using common cloud hosting and managed database services, with CI/CD supporting repeatable releases.

8 Engine availability heuristic

A mature set of frameworks, libraries, and services is available to implement Sortify without requiring the team to develop infrastructure from scratch.

- **Mobile framework:** React Native for cross-platform application development.
- **Backend framework:** FastAPI for REST APIs and backend services.
- **Database:** PostgreSQL or SQLite depending on the deployment stage.
- **ML framework:** PyTorch and established pretrained image-classification architectures.
- **Conversational AI:** an LLM API combined with retrieval-augmented generation over the waste knowledge base to power the chatbot.
- **Authentication:** token-based authentication with role-based authorization for users and administrators.
- **Testing:** suitable unit, integration, API, and frontend testing frameworks.
- **CI/CD:** automated build, test, and deployment pipelines.
- **Deployment:** cloud-hosted backend and database with a distributable mobile application.

We will use these established technologies to focus engineering effort on system integration, correctness, usability, testing, and maintainability rather than inventing new infrastructure.

9 Project scope and deliverables

9.1 Initial deliverable

- React Native application with the core navigation and user interface.
- User registration, login, and authentication.
- Image capture/upload workflow.
- FastAPI backend with image validation and prediction endpoint.
- Initial ML model integrated with the backend.
- Waste category and disposal-guidance database.
- Prediction result screen showing item, category, and confidence.
- Basic scan history.
- Backend unit/API tests and linting.

9.2 Subsequent deliverables

- Manual waste-item search and low-confidence fallback.
- Location-aware disposal guidance.
- User feedback and prediction correction.
- Personal statistics and activity dashboard.

- Educational waste-segregation content.
- Conversational assistant (chatbot) for natural-language disposal queries.
- Administrative dashboard.
- Waste-item and disposal-guideline management.
- Model-performance and prediction analytics.
- CI/CD and production deployment.

10 Risks and mitigations

- **ML misclassification:** use confidence thresholds, allow manual correction, and avoid presenting uncertain predictions as definitive.
- **Limited dataset coverage:** restrict the initial supported classes and clearly communicate the model's scope.
- **Variation in disposal rules:** associate disposal guidance with a defined location and maintain the information separately from the ML model.
- **Poor image quality:** validate uploaded images and provide clear instructions for retaking unclear photographs.
- **Data privacy:** minimize stored personal information and images, obtain appropriate user consent, and provide controls for user data.
- **Backend or network failure:** provide meaningful error states and allow manual access to locally available information where possible.
- **Scope expansion:** prioritize the core scan-to-guidance workflow and treat advanced features such as barcode recognition and on-device inference as optional extensions.
- **Chatbot reliability:** ground chatbot responses in the waste knowledge base via retrieval-augmented generation and restrict its scope to avoid hallucinated or unsafe disposal advice.

11 Summary

This project proposes **Sortify**, a mobile-first AI-powered waste identification and disposal guidance platform. The system combines a React Native mobile application, FastAPI backend, structured waste knowledge base, and lightweight image-classification model to help users move from an unidentified waste item to an actionable disposal recommendation.

The project meets the requirements of a Software Engineering project by emphasizing complete system development rather than ML research alone. It includes requirements-driven design, mobile and backend development, database management, authentication, API integration, testing, CI/CD, deployment, administrative workflows, and measurable evaluation. At the same time, the ML component provides a meaningful opportunity to investigate image classification, model evaluation, confidence handling, and feedback-driven improvement.

The initial scope will remain focused on a limited set of waste categories and supported items, allowing the team to build a reliable and maintainable system while leaving clear opportunities for future expansion.